

Malaysia EN/BM Cross-Language Consistency Test

Economic Report parallel pairs,
2014-2020 + 2022

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Framing

The C1 + C2 codebook pipeline was validated on US English documents. Malaysia has 38 BM-only documents in its ready corpus (out of 99 — primarily older Budget Speeches and Economic Reports). Before trusting LLM outputs on those, we need evidence that the pipeline produces equivalent fiscal reasoning regardless of input language.

This is a **consistency test, not a validity test**. We have no labels for Malaysia. EN outputs are treated as the reference because the codebooks crossed S3 gates on EN-only US data. Passing this test is necessary-but-not-sufficient for extending deployment to the **BM-only-document slice (38 of 99 ready documents)**; it does not gate the broader Malaysia deployment, and it is **not a substitute for expert agreement on the EN-side outputs** (Phase 2's actual success criteria: $\geq 80\%$ on C1, $\geq 70\%$ on C2 against expert review). Both gates are required for a complete Phase 2 validation. Failing this test would block deployment on BM-only documents absent a translation pre-processing step.

The test is self-contained: it slices `country_chunks` to Economic Reports with parallel EN+BM coverage, runs its own C1 $\rightarrow$ C2a $\rightarrow$ C2b chain on the slice, clusters near-duplicate measure names within each document, has Sonnet propose EN $\leftrightarrow$ BM cluster matches for human curation, then compares C2b motivation labels and signs on the curated matched pairs. Sonnet is used for matching

(different model family from the Haiku system under test); Haiku runs C1/C2a/C2b (the deployment model).

All reported rates carry bootstrap 95% confidence intervals ($n_{boot} = 1000$). No PASS/FAIL bar — interpretation belongs to the reader.

 Warning

This test only speaks to **modern, professionally-translated MoF prose** (Economic Reports 2014-2020, 2022). It does NOT speak to pre-2009 Budget Speech BM, older Economic Report BM (pre-2014), or crisis-booklet BM. Those are different vintages and registers; deployment to them requires separate evidence.

Scope: 10 pair years $\times$ 2 languages = 20 documents; 397 total chunks.

Pair years: 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2022, 2024, 2025

Level 1: Per-year act counts

Per year and language, count of distinct fiscal-measure clusters surfaced by the $C1 \rightarrow C2a \rightarrow$ clustering pipeline. A cluster is a group of `measure_name` strings within a single document with Jaro-Winkler distance ≤ 0.15 . Signed drift is $n_{bm} - n_{en}$; large absolute drift is a Level-1 finding regardless of Level 2 / Level 3 outcomes.

Table 1: Distinct measure clusters per year, by language

Year	n BM	n EN	Signed drift (BM – EN)	Drift %
2014	14	12	2	+14.3%
2015	5	7	–2	–28.6%
2016	4	6	–2	–33.3%
2017	8	4	4	+50.0%
2018	2	4	–2	–50.0%
2019	9	8	1	+11.1%
2020	4	15	–11	–73.3%
2022	5	5	0	+0.0%

Diagnostic: within-document name variance

The Level 1 drift counts depend entirely on the JW threshold (0.15) that groups near-duplicate `measure_name` strings into clusters within a single document. If JW fails to merge what is actually the same fiscal measure, the cluster count is inflated and apparent cross-language drift may be a clustering artifact rather than a real extraction asymmetry.

The figures below show, for each document, the pairwise JW distance matrix among the distinct `measure_name` strings C1 produced. Cells in the bottom band ($JW < 0.15$) were merged into one cluster by the pipeline. Cells in the middle band ($0.15 \leq JW < 0.30$) are the *near-miss* zone — same measure with verbose-vs-bare phrasing, swapped word order, or a parenthetical translation in one chunk but not another. Cells in the top band ($JW \geq 0.30$) are genuinely distinct measures.

Read each heatmap together with its numbered measure list (printed below it). Off-diagonal cells just above the 0.15 threshold linking semantically-identical names are the failure cases that drive over-fragmentation.

2014 EN — 12 distinct C1 measure names

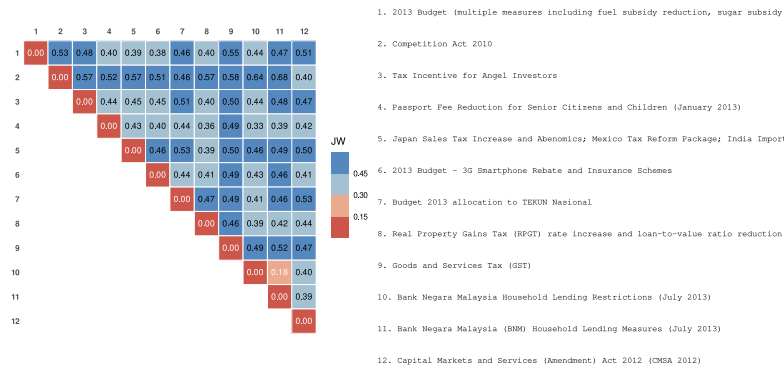


Figure 1: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2014 BM — 14 distinct C1 measure names

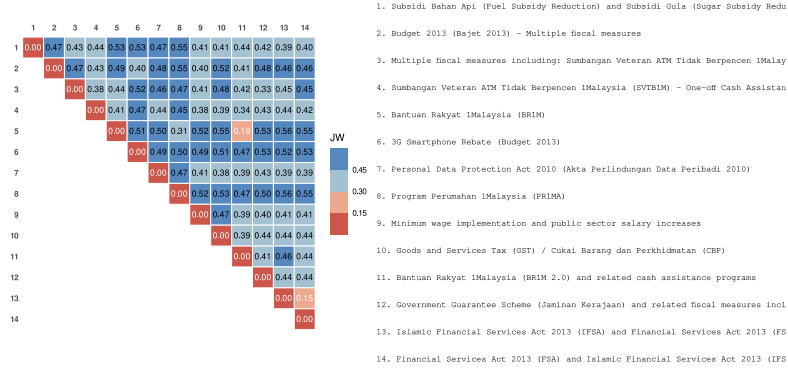


Figure 2: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2015 EN — 8 distinct C1 measure names

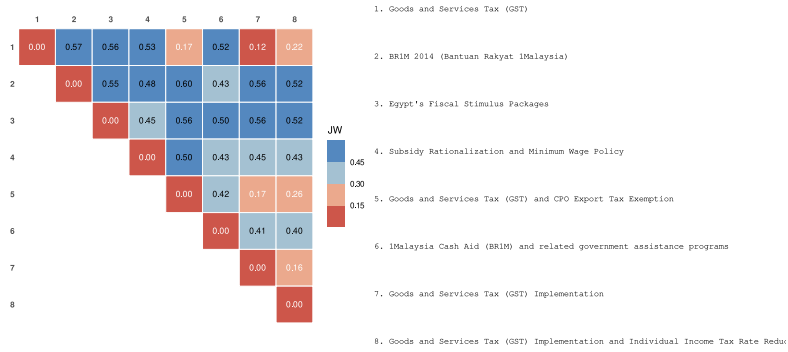


Figure 3: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2015 BM — 9 distinct C1 measure names

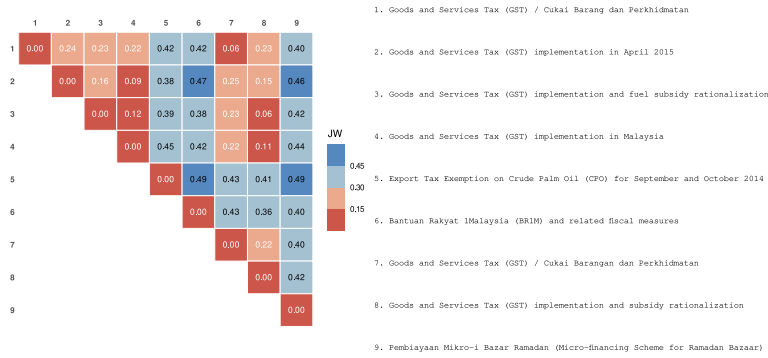


Figure 4: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2016 EN — 6 distinct C1 measure names

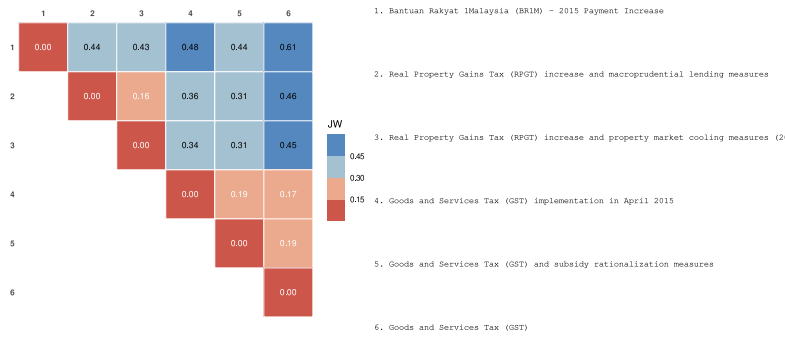


Figure 5: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2016 BM — 6 distinct C1 measure names

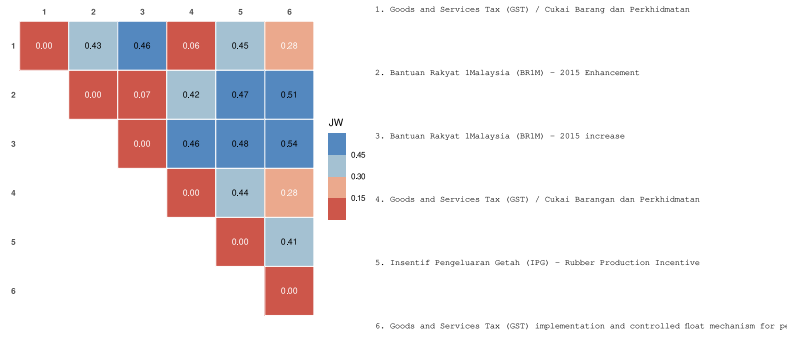


Figure 6: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2017 EN — 6 distinct C1 measure names

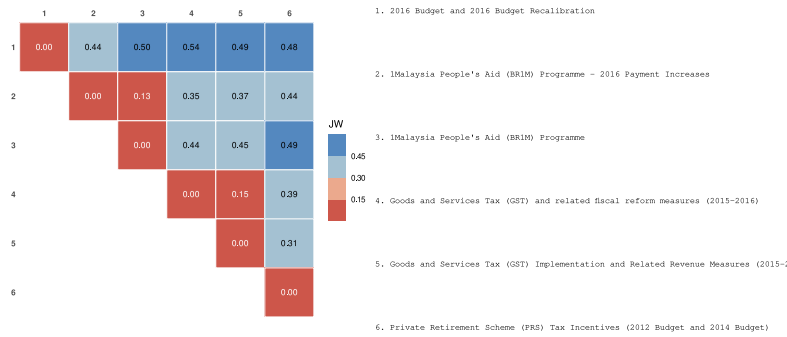


Figure 7: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2017 BM — 9 distinct C1 measure names

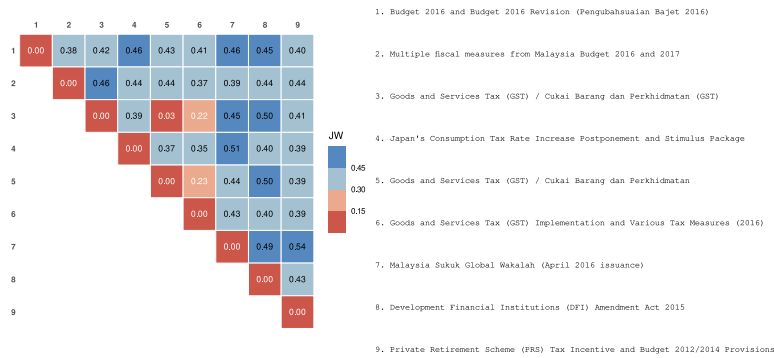


Figure 8: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2018 EN — 4 distinct C1 measure names

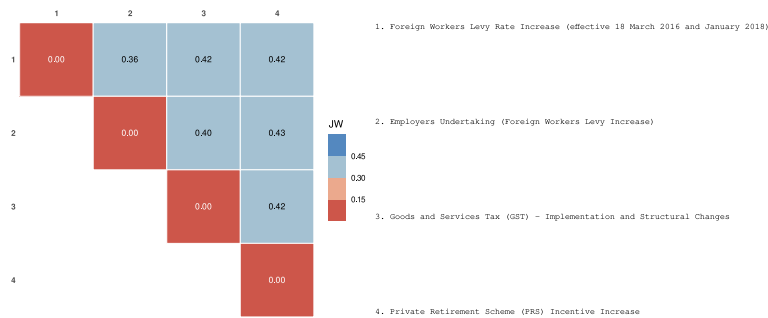


Figure 9: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2018 BM — 2 distinct C1 measure names

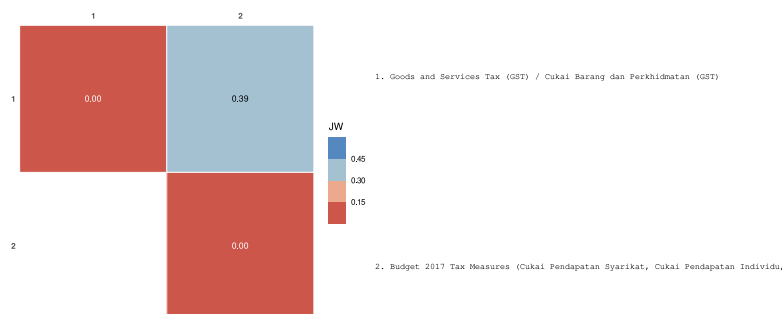


Figure 10: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells (0.15-0.30) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2019 EN — 9 distinct C1 measure names

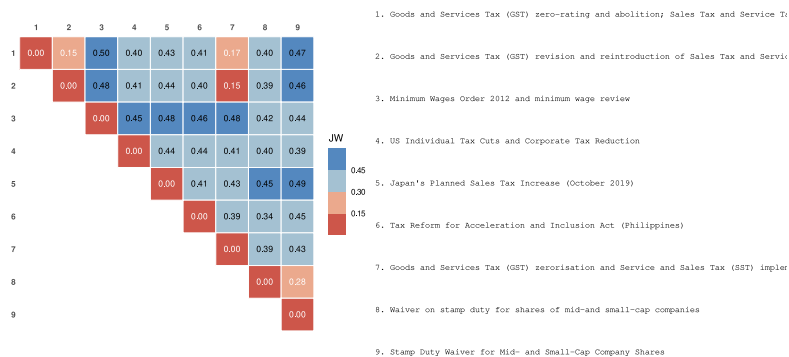


Figure 11: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells (0.15-0.30) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2019 BM — 10 distinct C1 measure names

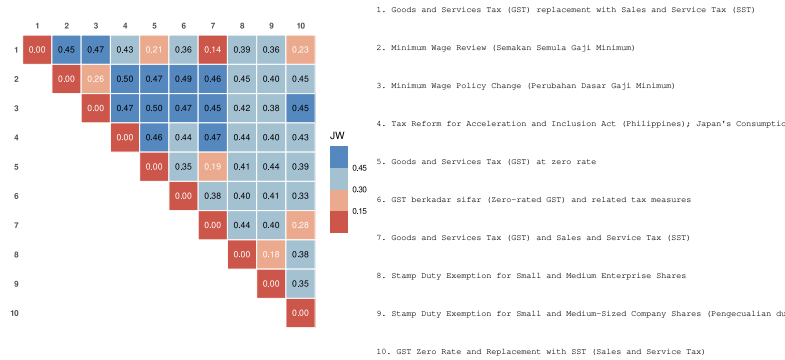


Figure 12: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells (JW < 0.15) were merged; orange cells (0.15-0.30) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2020 EN — 20 distinct C1 measure names

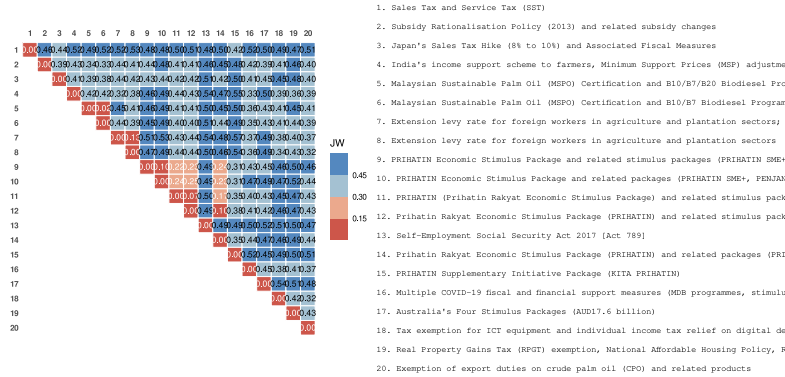


Figure 13: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells (JW < 0.15) were merged; orange cells (0.15-0.30) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2020 BM — 4 distinct C1 measure names

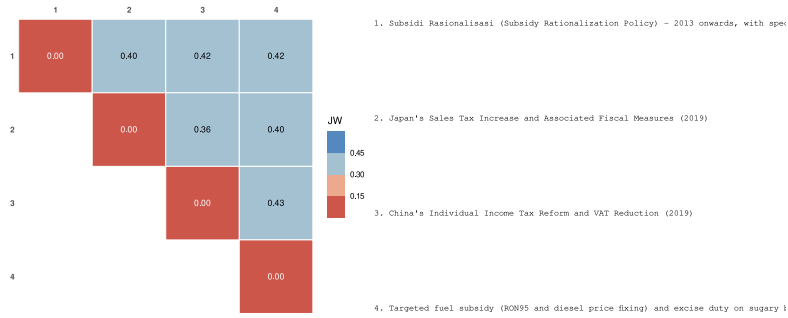


Figure 14: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2022 EN — 5 distinct C1 measure names

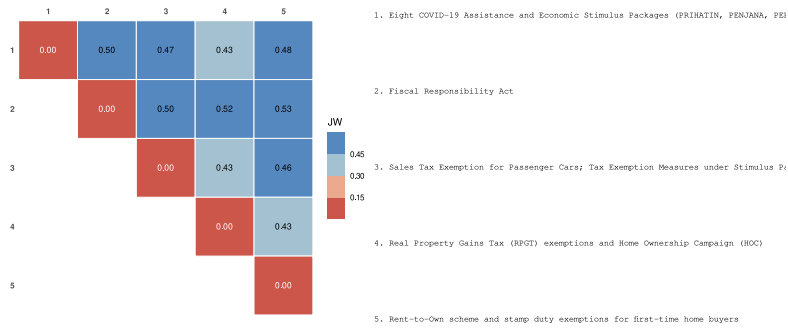


Figure 15: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

2022 BM — 6 distinct C1 measure names

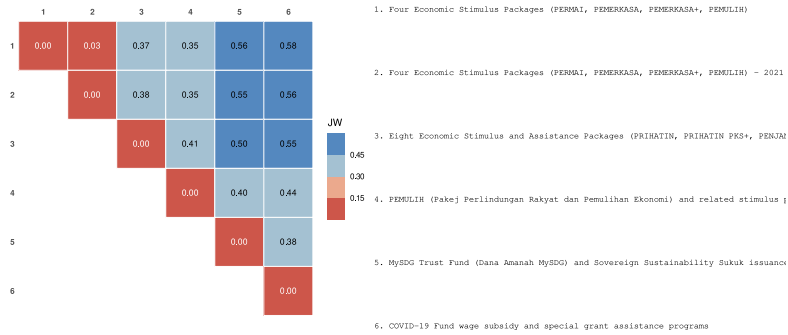


Figure 16: Pairwise Jaro-Winkler distance among distinct C1 measure names within each document. Red cells ($JW < 0.15$) were merged; orange cells ($0.15-0.30$) are near-misses that did not merge but plausibly should have; blue cells (≥ 0.30) are distinct.

Threshold sensitivity of Level 1 drift

The JW threshold (0.15) is the single tuning parameter behind Level 1. If apparent drift is sensitive to relaxing this threshold, the drift figure is a clustering artifact rather than evidence about cross-language model behaviour. The table below re-runs the same single-linkage hierarchical clustering on the same C1 `measure_name` strings at progressively looser thresholds, holding everything else fixed.

Table 2: Signed drift (BM – EN) at varying JW thresholds. Production setting is $JW \leq 0.15$ (leftmost column). A drift figure that collapses toward zero at relaxed thresholds was a clustering artifact; a drift figure that persists is a real extraction asymmetry.

Year	JW $\leq$ 0.15	JW $\leq$ 0.20	JW $\leq$ 0.25	JW $\leq$ 0.30	JW $\leq$ 0.35
2014	+2 (12/14)	+1 (11/12)	+1 (11/12)	+1 (11/12)	–1 (10/9)
2015	–2 (7/5)	+0 (5/5)	–1 (5/4)	–1 (5/4)	–1 (5/4)
2016	–2 (6/4)	+1 (3/4)	+1 (3/4)	+0 (3/3)	+1 (2/3)
2017	+4 (4/8)	+4 (4/8)	+3 (4/7)	+3 (4/7)	+4 (3/7)
2018	–2 (4/2)	–2 (4/2)	–2 (4/2)	–2 (4/2)	–2 (4/2)
2019	+1 (8/9)	+0 (7/7)	–1 (7/6)	–1 (6/5)	–2 (5/3)
2020	–11 (15/4)	–11 (15/4)	–10 (14/4)	–10 (14/4)	–3 (7/4)
2022	+0 (5/5)	+0 (5/5)	+0 (5/5)	+0 (5/5)	–1 (5/4)

Each cell shows `drift (n_EN/n_BM)`. Two patterns to read:

- **Clustering-driven drift (years that collapse toward 0 as threshold relaxes):** 2015 and 2016 are textbook artifacts — both drop to zero or near-zero at $JW \leq 0.20$. 2014 and 2019 are partial artifacts.
- **Real extraction asymmetry (years robust to threshold):** 2017, 2018, and 2020 show drift that survives even at $JW \leq 0.30$. These are not clustering failures and warrant a separate explanation. Cross-referencing the within-doc heatmaps above: 2020 EN contains several non-Malaysian comparator clusters (Japan, India, Australia stimulus packages), 2017 BM shows the same contamination on the BM side (Japan, Sukuk issuance), and 2018 BM extracted only 2 clusters from a full Economic Report — a suspiciously low recall figure unrelated to clustering. These residuals are tractable but require upstream fixes (country-of-enactment tagging in C1; investigation of the 2018 BM recall) rather than JW-threshold tuning.

Level 2: Pairwise matching

Per year, Sonnet proposes candidate $EN \leftrightarrow BM$ cluster matches with confidence labels. A human reviewer reads the candidates CSV and produces a curated CSV with the matches they

accept. This section’s match rate is $n_{\text{curated}} / n_{\text{proposed}}$; $n_{\text{human_overrides}}$ is the count of LLM-proposed matches the human rejected (a matcher-bias diagnostic).

Table 3: Curation status and match rate

Proposed by LLM	Curated by human	Match rate	Human overrides
28	27	96.4% [89.3%, 100.0%]	1

This iteration accepted 27 of 28 LLM-proposed pairs and added zero matches manually. The single rejection is the 2014 passport-fee/omnibus row (EN cluster 4 $\leftrightarrow$ BM cluster 3) where Sonnet paired a single-measure EN cluster with a multi-measure BM cluster spanning four unrelated programmes. The 96.4% match rate is a high-confidence Sonnet performance number on a small sample.

Zero human-added matches is itself an interpretive finding rather than evidence of complete Sonnet recall. Several `bm_unmatched` clusters share semantic content with EN omnibus clusters. For example, BM 1 in 2014 (`Subsidi Bahan Api / Subsidi Gula`) overlaps thematically with EN 1 (the 2013 Budget omnibus, which mentions fuel and sugar subsidy reduction inside its prose), but EN 1 was already paired with BM 2 (the BM 2013 Budget omnibus). The decision not to manually pair these fragments reflects an upstream clustering asymmetry: BM tends to fragment a single fiscal act into multiple measure-name clusters (subsidies broken out separately from the budget envelope), while EN tends to keep them aggregated. There is no clean 1-to-1 manual match available when the underlying clusters are not 1-to-1 acts.

Candidate matches per year

Table 4: LLM-proposed candidate matches (review context)

Year	EN	canoni- cal	BM	canoni- cal	Confidence	Status
2014		—	Subsidi han	Ba- Api (Fuel Sub- sidy Re- duction) and Subsidi Gula (Sugar Subsidy Re- duction)	medium	bm_unmatched
2014		—	Sumban- gan Veteran ATM Tidak Berpencen 1Malaysia (SVTB1M) - One-off Cash Assis- tance		medium	bm_unmatched
2014		—	Bantuan Rakyat 1Malaysia (BR1M)		medium	bm_unmatched
2014		—	Personal Data Pro- tection Act 2010 (Akta Perlindun- gan Data Peribadi 2010)		low	bm_unmatched
2014		—	Program Perumahan 1Malaysia (PR1MA)		low	bm_unmatched
2014		—	Minimum wage imple- mentation and pub- lic sector salary in- creases		low	bm_unmatched

Cluster lists per year (review aid)

For each year, the EN and BM cluster lists side-by-side, with canonical names and the count of contributing chunks. Use this when curating the candidates CSV to spot LLM-missed matches.

2014

EN

ID	Canonical name
1	2013 Budget (multiple measures including fuel subsidy reduction, sugar subsidy reduction, cash transfers, and various investment allocations)
2	Competition Act 2010
3	Tax Incentive for Angel Investors
4	Passport Fee Reduction for Senior Citizens and Children (January 2013)
5	Japan Sales Tax Increase and Abenomics; Mexico Tax Reform Package; India Import Tax on Gold; Brazil Fiscal Stimulus; Russia Stimulus Package
6	2013 Budget - 3G Smartphone Rebate and Insurance Schemes
7	Budget 2013 allocation to TEKUN Nasional
8	Real Property Gains Tax (RPGT) rate increase and loan-to-value ratio reduction
9	Goods and Services Tax (GST)

10 Bank Negara Malaysia Household Goods and Services Tax (GST) 2013)

BM

ID	Canonical name
1	Subsidi Bahan Api (Fuel Subsidy Reduction) and Subsidi Gula (Sugar Subsidy Reduction)
2	Budget 2013 (Bajet 2013) - Multiple fiscal measures
3	Multiple fiscal measures including: Sumbangan Veteran ATM Tidak Berpencen 1Malaysia (SVTB1M), Passport Fee Reduction for Seniors, ATM Service Scheme Review with salary increases, and Group Insurance Protection Scheme
4	Sumbangan Veteran ATM Tidak Berpencen 1Malaysia (SVTB1M) - One-off Cash Assistance
5	Bantuan Rakyat 1Malaysia (BR1M)
6	3G Smartphone Rebate (Budget 2013)
7	Personal Data Protection Act 2010 (Akta Perlindungan Data Peribadi 2010)
8	Program Perumahan 1Malaysia (PR1MA)
9	Minimum wage im-

10 ~~Implementation of and public sector salary increases, including fee imposition on guarantees (2013, onwards), and~~

2015

EN

ID	Canonical name
1	Goods and Services Tax (GST) Implementation
2	BR1M 2014 (Bantuan Rakyat 1Malaysia)
3	Egypt's Fiscal Stimulus Packages
4	Subsidy Rationalization and Minimum Wage Policy
5	Goods and Services Tax (GST) and CPO Export Tax Exemption
6	1Malaysia Cash Aid (BR1M) and related government assistance programs
7	Goods and Services Tax (GST) Implementation and Individual Income Tax Rate Reduction

BM

ID	Canonical name
1	Goods and Services Tax (GST) / Cukai Barangan dan Perkhidmatan
2	Goods and Services Tax (GST) implementation and fuel subsidy rationalization
3	Export Tax Exemption on Crude Palm Oil (CPO) for September and October 2014
4	Bantuan Rakyat 1Malaysia (BR1M) and related fiscal measures
5	Pembiayaan Mikro-i Bazar Ramadan (Micro-financing Scheme for Ramadan Bazaar)

2016**EN**

ID	Canonical name
1	Bantuan Rakyat 1Malaysia (BR1M) - 2015 Payment Increase
2	Real Property Gains Tax (RPGT) increase and macroprudential lending measures
3	Real Property Gains Tax (RPGT) increase and property market cooling measures (2014 Budget)
4	Goods and Services Tax (GST) implementation in April 2015
5	Goods and Services Tax (GST) and subsidy rationalization measures
6	Goods and Services Tax (GST)

BM

ID	Canonical name
1	Goods and Services Tax (GST) / Cukai Barangan dan Perkhidmatan
2	Bantuan Rakyat 1Malaysia (BR1M) - 2015 Enhancement
3	Insentif Pengeluaran Getah (IPG) - Rubber Production Incentive
4	Goods and Services Tax (GST) implementation and controlled float mechanism for petroleum pricing

2017**EN**

ID	Canonical name
1	2016 Budget and 2016 Budget Recalibration
2	1Malaysia People's Aid (BR1M) Programme - 2016 Payment Increases
3	Goods and Services Tax (GST) Implementation and Related Revenue Measures (2015-2016)
4	Private Retirement Scheme (PRS) Tax Incentives (2012 Budget and 2014 Budget)

BM

ID	Canonical name
1	Budget 2016 and Budget 2016 Revision (Pengubahsuaian Bajet 2016)
2	Multiple fiscal measures from Malaysia Budget 2016 and 2017
3	Goods and Services Tax (GST) / Cukai Barang dan Perkhidmatan (GST)
4	Japan's Consumption Tax Rate Increase Postponement and Stimulus Package
5	Goods and Services Tax (GST) Implementation and Various Tax Measures (2016)
6	Malaysia Sukuk Global Wakalah (April 2016 issuance)
7	Development Financial Institutions (DFI) Amendment Act 2015
8	Private Retirement Scheme (PRS) Tax Incentive and Budget 2012/2014 Provisions

2018**EN**

ID	Canonical name
1	Foreign Workers Levy Rate Increase (effective 18 March 2016 and January 2018)
2	Employers Undertaking (Foreign Workers Levy Increase)
3	Goods and Services Tax (GST) - Implementation and Structural Changes
4	Private Retirement Scheme (PRS) Incentive Increase

BM

ID	Canonical name
1	Goods and Services Tax (GST) / Cukai Barang dan Perkhidmatan (GST)
2	Budget 2017 Tax Measures (Cukai Pendapatan Syarikat, Cukai Pendapatan Individu, dan Insentif Cukai)

2019

EN

ID	Canonical name
1	Goods and Services Tax (GST) zero-rating and abolition; Sales Tax and Service Tax (SST) reintroduction
2	Goods and Services Tax (GST) revision and reintroduction of Sales Tax and Service Tax (SST)
3	Minimum Wages Order 2012 and minimum wage review
4	US Individual Tax Cuts and Corporate Tax Reduction
5	Japan's Planned Sales Tax Increase (October 2019)
6	Tax Reform for Acceleration and Inclusion Act (Philippines)
7	Waiver on stamp duty for shares of mid-and small-cap companies
8	Stamp Duty Waiver for Mid- and Small-Cap Company Shares

BM

ID	Canonical name
1	Goods and Services Tax (GST) replacement with Sales and Service Tax (SST)
2	Minimum Wage Review (Semakan Semula Gaji Minimum)
3	Minimum Wage Policy Change (Perubahan Dasar Gaji Minimum)
4	Tax Reform for Acceleration and Inclusion Act (Philippines); Japan's Consumption Tax Increase; Australia's Income Tax Reduction; Korea's Minimum Wage Increase
5	Goods and Services Tax (GST) at zero rate
6	GST berkadar sifar (Zero-rated GST) and related tax measures
7	Stamp Duty Exemption for Small and Medium Enterprise Shares
8	Stamp Duty Exemption for Small and Medium-Sized Company Shares (Pengecualian duti setem)
9	GST Zero Rate and Replacement with SST (Sales and Service Tax)

2020

EN

ID	Canonical name
1	Sales Tax and Service Tax (SST)
2	Subsidy Rationalisation Policy (2013) and related subsidy changes
3	Japan's Sales Tax Hike (8% to 10%) and Associated Fiscal Measures
4	India's income support scheme to farmers, Minimum Support Prices (MSP) adjustment, and tax relief for taxpayers earning less than INR500,000 annually
5	Malaysian Sustainable Palm Oil (MSPO) Certification and B10/B7/B20 Biodiesel Programmes
6	Extension levy rate for foreign workers in agriculture and plantation sectors; Sugar tax; Departure levy; Targeted fuel subsidy
7	PRIHATIN Economic Stimulus Package and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)

~~8 PRIHATIN (Prihatin~~
~~10 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~11 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~12 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
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~~31 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~32 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~33 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~34 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~35 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~36 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~37 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~38 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~39 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~40 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
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~~45 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~46 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~47 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~48 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~49 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~
~~50 2018-2019 Economic Stimulus Package) and related stimulus packages (PRIHATIN SME+, PENJANA, KITA PRIHATIN)~~

BM

ID	Canonical name
1	Subsidi Rasionalisasi (Subsidy Rationalization Policy) - 2013 onwards, with specific 2015 changes
2	Japan's Sales Tax Increase and Associated Fiscal Measures (2019)
3	China's Individual Income Tax Reform and VAT Reduction (2019)
4	Targeted fuel subsidy (RON95 and diesel price fixing) and excise duty on sugary beverages

2022

EN

ID	Canonical name
1	Eight COVID-19 Assistance and Economic Stimulus Packages (PRIHATIN, PENJANA, PERMAI, PEMERKASA, PEMULIH, and related programs)
2	Fiscal Responsibility Act
3	Sales Tax Exemption for Passenger Cars; Tax Exemption Measures under Stimulus Packages; Special Tax Relief for Digital Devices; Real Property Gains Tax (RPGT) Exemptions; Electricity Bill Discounts under PEMULIH
4	Real Property Gains Tax (RPGT) exemptions and Home Ownership Campaign (HOC)
5	Rent-to-Own scheme and stamp duty exemptions for first-time home buyers

BM

ID	Canonical name
1	Four Economic Stimulus Packages (PERMAI, PEMERKASA, PEMERKASA+, PEMULIH) - 2021
2	Eight Economic Stimulus and Assistance Packages (PRIHATIN, PRIHATIN PKS+, PENJANA, KITA PRIHATIN, PERMAI, PEMERKASA, PEMERKASA+, PEMULIH) totaling RM530 billion
3	PEMULIH (Pakej Perlindungan Rakyat dan Pemulihan Ekonomi) and related stimulus packages (PERMAI, PEMERKASA, PEMERKASA+)
4	MySDG Trust Fund (Dana Amanah MySDG) and Sovereign Sustainability Sukuk issuances
5	COVID-19 Fund wage subsidy and special grant assistance programs

Level 3: Classification agreement

Conditional on curated matched pairs, do C2b motivation labels and signs agree between the EN-side and BM-side classifications of the same act?

Table 5: Headline agreement rates on matched pairs

Metric	Rate	n pairs
Label agreement	63.0% [44.4%, 81.5%]	27
Sign agreement	74.1% [55.6%, 88.9%]	27

Three of the 10 disagreement rows on this run are curation artifacts where the matched cluster pair is not actually the same fiscal act:

- **Pair 19 (2019):** EN = Japan sales tax increase; BM = Tax Reform for Acceleration and Inclusion Act (Philippines). Two foreign comparators that should both have been filtered out of the Malaysian corpus upstream.
- **Pair 26 (2022):** EN = Fiscal Responsibility Act; BM = MySDG Trust Fund. Different acts.
- **Pair 27 (2022):** EN = Sales Tax Exemption for Passenger Cars; BM = PEMULIH (COVID assistance). Different policies, same year.

Reclassifying those three pairs as non-matches moves the headline to 70.8% label / 70.8% sign agreement on $n = 24$, just touching the Phase 2 C2 expert-agreement floor ($\geq 70\%$) but with a CI still too wide for this test to resolve readiness on its own.

The Phase-2 readiness question is also confounded by an upstream issue independent of curation. The $C1 \rightarrow C2a \rightarrow$ within-doc clustering chain produces fragmented clusters that are not 1-to-1 with fiscal acts (see the within-doc heatmaps above). When Sonnet matches a BM omnibus cluster to an EN omnibus cluster, the BM omnibus has often had its deficit-driven instruments (subsidy cuts, cash transfers, levies) split off into separate clusters that go unmatched, while the EN side concatenates the full act. C2b is therefore reading asymmetric evidence on the two sides: the EN side sees a complete picture of the act, the BM side sees a fragment with the deficit-driven content removed. Apparent cross-language disagreement is partly a consequence of this asymmetric input rather than evidence about cross-language model behaviour.

The robust finding underneath the noise is that Haiku’s motivation classification lands on the same label across languages 63% of the time *despite* the asymmetric inputs. With clean cluster matches, the rate is likely higher. This is a robustness statement about C2b useful for deployment confidence on its own terms; the headline agreement rates as currently computed are better read as an upper

bound on cross-language noise than as a measurement of cross-language model behaviour.

Confusion matrix (EN labels vs BM labels)

Table 6: Motivation label confusion matrix

EN BM	BMBM: DEFICIT_DRIVEN	DEFICIT_DRIVEN	LONG_RUN	NONENDING_DRIVEN
COUN- TER- CYCLI- CAL	4	0	2	0
DEFICIT_DRIVEN	0	4	4	0
LONG_RUN	1	0	9	2
NA	0	1	0	0

Disagreement rows

When EN and BM disagree, the most informative rows are those where both sides expressed high confidence — those are the genuine model-behaviour disagreements rather than borderline calls.

Direction of disagreement

The value of this test is in the *pattern* of disagreement, not the headline rate. Three patterns surface in the confusion matrix for this run:

- **BM-side LONG_RUN over-representation.** The dominant off-diagonal cell is DEFICIT_DRIVEN (EN) → LONG_RUN (BM) at 4 pairs, with two secondary cells in the same direction: COUNTERCYCLICAL (EN) → LONG_RUN (BM) at 2 pairs, and LONG_RUN (EN) → SPENDING_DRIVEN (BM) at 2 pairs. The BM column total for LONG_RUN is 15 vs an EN row total of 12. The directional reading is that BM reaches for the long-run/structural label more readily than EN.
- **Plausibly a clustering artifact, not a model bias.** If the BM omnibus cluster that gets matched to EN systematically excludes the deficit-driven instruments (subsidies, cash transfers) split off into separate clusters, the BM-side evidence base is depleted of deficit-driven framing and reads more structural by construction. Testing this requires pulling the per-side evidence for the 4 DEFICIT_DRIVEN→LONG_RUN pairs and checking whether the unmatched BM fragments in those years contain the deficit-driven content. If they do, the directional finding dissolves into clustering asymmetry; if they don't, a BM-side model bias survives the artifact.
- **Sign-only disagreements (3 pairs) are cleaner signal.** Pair 2 (3G smartphone rebate, both LONG_RUN, EN +, BM -), pair 7 (subsidy rationalisation, both DEFICIT_DRIVEN, EN -, BM +), and pair 27 (likely curation error, see above). Two are likely BM-side direction-reasoning failures: subsidy reduction is fiscally contractionary, so EN's - is correct on pair 7; a rebate is expansionary, so EN's + is correct on pair 2.

The `both_high_confidence` column in the disagreement table below sorts as a no-op because of a vectorisation bug in `malay_consistency.R:752 (identical())` on a vector returns `FALSE` for any length > 1). Confidence asymmetry between EN and BM is therefore not yet readable from this notebook.

i Methodological caveat: translation, not independent source

The BM Economic Reports are MoF translations of the EN versions, not independent fiscal-policy statements. This has two consequences for how to read Level 3:

- **Correlated errors are invisible.** If Haiku misreads a motivation passage the same way in both languages (because the BM translation preserved the misleading phrasing), this test will record agreement and the error will not appear here. Catching correlated errors requires expert validation on the EN side.
- **Some disagreements are translation artifacts.** When EN and BM C2b labels diverge, the cause might be the translator's word choice rather than the model's cross-language reasoning. The motivation language R&R rely on ("the actual words used by policymakers") only fully applies to the EN side; BM is a mediated rendering.

The Sonnet matcher proposes EN ↔ BM matches but cannot add matches for clusters it flags as `bm_unmatched`. When reviewing the candidates CSV, also scan for BM clusters tagged `bm_unmatched` whose canonical name looks plausibly related to an EN cluster — those are matches the matcher missed and the human should manually convert by editing the CSV.

Confidence asymmetry:

- EN high-confidence rate: 77% (20/27)
- BM high-confidence rate: 74% (20/27)

Iteration synthesis

This iteration's primary deliverable is the diagnosis of where the pipeline breaks, not the headline agreement rates. Three findings, in order of load-bearing:

1. **Within-doc measure-name clustering is the upstream bottleneck.** The $JW \leq 0.15$ cluster step is producing fragments rather than acts. The within-doc heatmaps make this directly visible: many near-miss cells in the 0.15-0.30 band, particularly on the BM side. Fragmentation propagates downstream. Cross-language matching is overspecified at the cluster level rather than the act level, and C2b reads asymmetric evidence on the two sides because BM tends to fragment what EN aggregates.

The Level 2 candidate quality and Level 3 disagreement patterns are partly artifacts of this upstream failure rather than evidence about cross-language model behaviour.

2. **C2b classification is robust to degraded inputs.** Despite asymmetric evidence on the two sides, Haiku's motivation labels agree 63% of the time and signs agree 74% of the time. With clean cluster matches, the rate is likely higher. This is a robustness statement about the deployed model, useful for cross-language deployment confidence on its own terms.
3. **The Phase-2 readiness question is unresolved.** The agreement rates as currently computed are an upper bound on cross-language noise because clustering noise is folded in, so they cannot resolve whether the BM-only deployment clears the $\geq 70\%$ expert-agreement floor on C2. The test becomes diagnostic once clustering is fixed.

The cheapest directional test of finding (1) is to re-run Levels 2 and 3 at relaxed JW thresholds (0.25, 0.30) using existing infrastructure. The threshold-sensitivity table earlier in this notebook shows the within-doc clustering effect on Level 1 drift; an analogous sweep at Level 3 would tell us whether the disagreement pattern is clustering-driven (rates climb when threshold relaxes) or model-driven (rates flat). The one-time re-curation cost is real but bounded: cluster IDs change across re-clustering runs, so the curated CSV needs to be redone, but a relaxed-threshold run produces fewer clusters per year and the second curation pass is faster than the first.

A more architectural alternative is to push canonicalisation into C1 itself rather than papering over name variance downstream. Two downstream variants are also available: tightening the JW threshold parameter (cheap, mechanical) and replacing the JW step with a Sonnet-based dedup pass between `c2a_evidence` and `clusters` (more expensive, robust to phrasing variance). The cheap threshold sweep above informs which is worth building.

Implications for deployment scope

This test, conditional on its results, speaks to a narrow population of BM documents. Reading the test out for deployment scope:

- **Speaks to:** professional MoF translations of modern Economic Reports (2014+). The translations are commissioned alongside the EN versions and are expected to be high-quality.
- **Does NOT speak to:** older Economic Report BM (1995-2013, 19 documents — these are MoF Bahasa originals, not parallel

translations), **all Budget Speech BM (9 documents spanning 2000-2008, 2012, 2015 — none have EN parallels regardless of year, and the speech register differs structurally from Economic Report prose)**, or crisis-booklet BM (4 documents — branded marketing register).

The 2024 and 2025 EN+BM ER pairs exist on disk but fall outside the current Malaysia pilot window (`pilot_year_max = 2022`). Extending the test to those years requires only an update to `build_country_configs()`; the consistency-test sub-pipeline auto-includes any new pair years from `MANIFEST.csv`.